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Assignment 7

Routing using OSPF (Link State Routing) in Cisco Packet Tracer

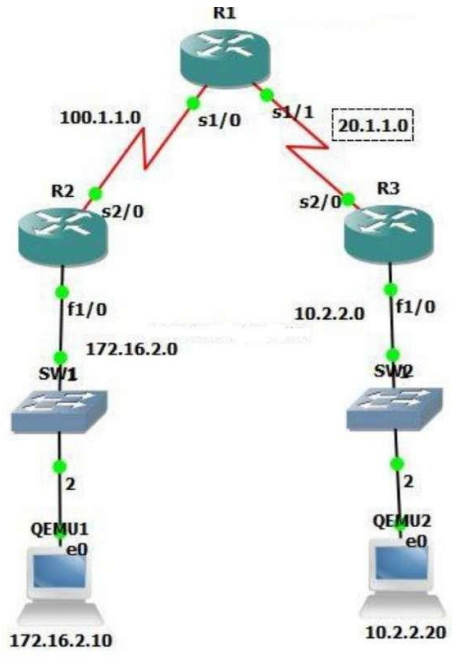
Objective

- To understand the **Link State Routing algorithm (OSPF)**
- To configure and verify **OSPF routing** in a network
- To analyze how **Shortest Path First (SPF)** algorithm works

Problem Statement

Design and implement the given network topology with three routers (R1, R2, R3), two LANs, and PCs.

In addition to configuration, **analyze route selection using OSPF cost and SPF algorithm**.



Network Topology & IP Addressing

- 3 Routers: R1, R2, R3
- 2 LANs
- Serial links between routers
- PCs connected to R2 and R3

Tasks

◆ Part A: Basic Configuration

1. Create the network topology in Cisco Packet Tracer.
2. Configure IP addresses on all router interfaces and PCs.
3. Verify connectivity between directly connected devices

Part B: OSPF Configuration

1. Enable OSPF on all routers:

router ospf 1

2. Configure router IDs manually (recommended):

router-id 1.1.1.1

3. Advertise networks using:

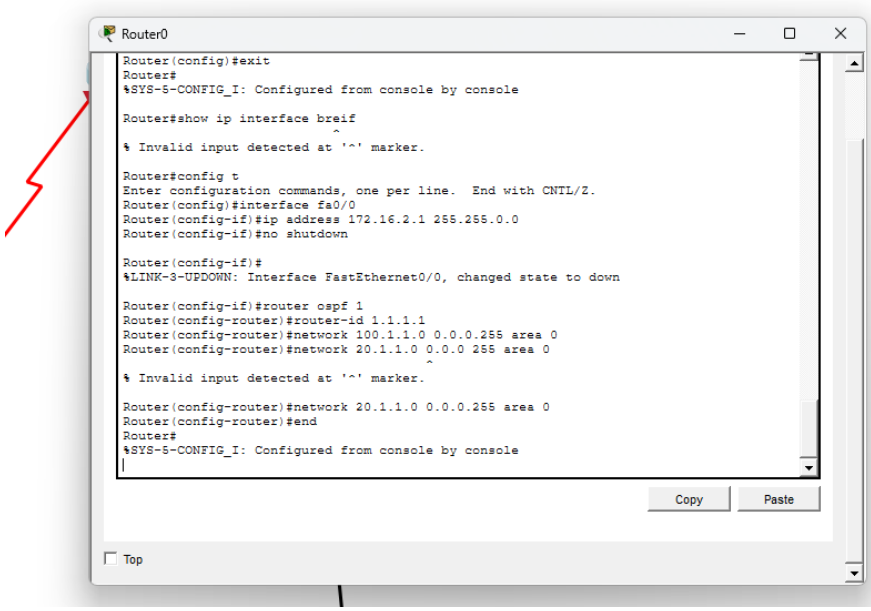
network <network-address> <wildcard-mask> area 0

4. Ensure all routers are in **Area 0 (Backbone Area)**

◆ Sample Configuration

R1

router ospf 1
router-id 1.1.1.1
network 100.1.1.0 0.0.0.255 area 0
network 20.1.1.0 0.0.0.255 area 0



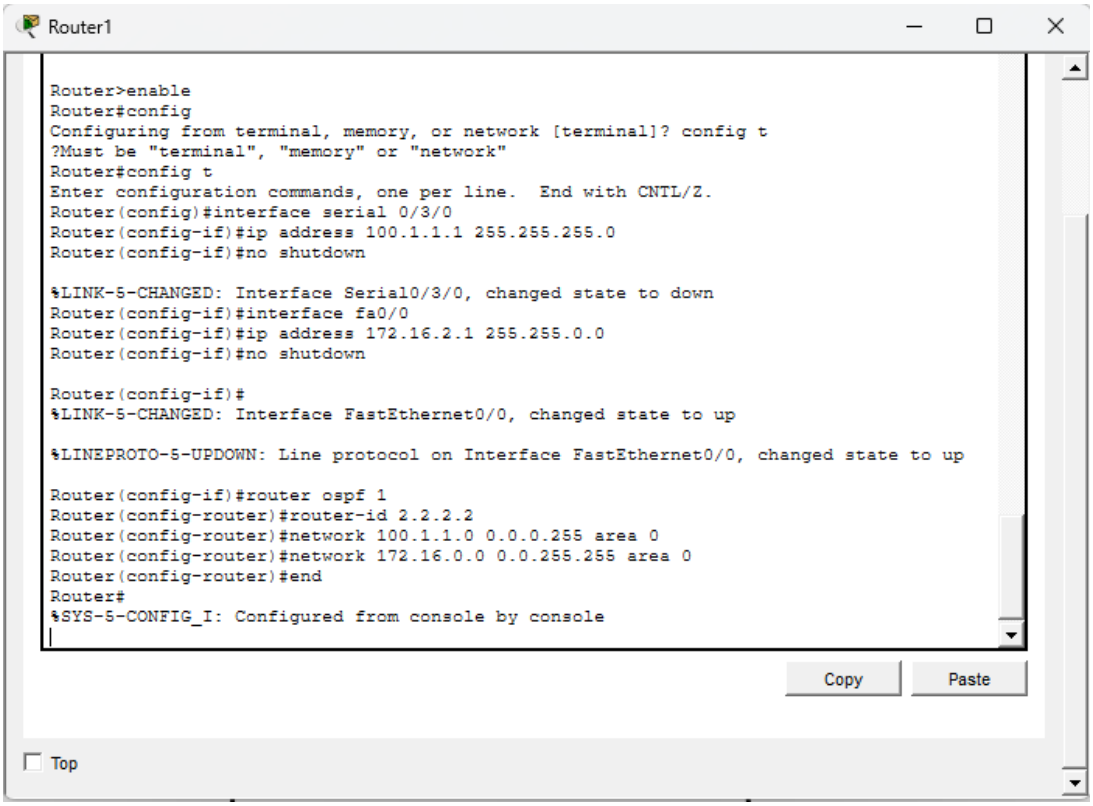
R2

router ospf 1

router-id 2.2.2.2

network 100.1.1.0 0.0.0.255 area 0

network 172.16.0.0 0.0.255.255 area 0



R3

router ospf 1

router-id 3.3.3.3

network 20.1.1.0 0.0.0.255 area 0

network 10.2.2.0 0.0.0.255 area 0

Router2

```
Press RETURN to get started!



Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 20.1.1.2 255.255.255.0
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#interface fa0/0
Router(config-if)#ip address 10.2.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#router-id 3.3.3.3
Router(config-router)#network 20.1.1.0 0.0.0.255 area 0
Router(config-router)#network 10.2.2.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

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Top

Part C: Link State (OSPF) Analysis (NEW)

Step 1: Neighbor Discovery

- Use:

show ip ospf neighbor

- Identify OSPF neighbors for each router

```
Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           0   FULL/  -        00:00:30    100.1.1.1       Serial0/3/0
```

Step 2: Link State Database (LSDB)

- Use:

show ip ospf database

```
Router#show ip ospf database
            OSPF Router with ID (1.1.1.1) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         308         0x80000003 0x005c89 2
2.2.2.2         2.2.2.2         308         0x80000003 0x008292 3
Router#
```

- Observe LSAs shared between routers

Step 3: SPF Calculation (Conceptual)**

Students must:

- Draw network graph
- Assign **cost to each link** (default or manual)
- Apply **Shortest Path First (Dijkstra Algorithm)**

Part D: Verification Commands

- Check routing table:

show ip route

-
```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.1.1.0/24 is directly connected, Serial0/3/0
L       100.1.1.2/32 is directly connected, Serial0/3/0
O    172.16.0.0/16 [110/65] via 100.1.1.1, 00:05:58, Serial0/3/0

Router#
```

Part E: Simulation & Testing

1. Ping from PC1 → PC2

2.
```
PC0

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.2.1

Pinging 172.16.2.1 with 32 bytes of data:

Reply from 172.16.2.1: bytes=32 time<1ms TTL=255
Reply from 172.16.2.1: bytes=32 time=3ms TTL=255
Reply from 172.16.2.1: bytes=32 time<1ms TTL=255
Reply from 172.16.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.16.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>

Top
```

3. Use:

traceroute

4.

```
C:\>tracert 10.2.2.20

Tracing route to 10.2.2.20 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      172.16.2.1
  2   0 ms      *         0 ms      172.16.2.1
```

5. Change bandwidth on one link:

bandwidth 64

6. Observe route change based on cost

Additional Analytical Task (NEW)

Fill the table:

Router	**Destination**	**Path**	**Cost**
R1	10.2.2.0	R1→R3	64